

Experiment B Executive Summary

Solar LSTM | Plant2 -> Plant1 | Final Scientific Closure Summary

Project	LSTM 研究與實驗－SolarEnergy 太陽能發電預測（智慧能源）
Experiment	Experiment B：Plant2 -> Plant1
Document date	2026-09-15
Scientific status	SEALED / CLOSED
Model-development rule	NO MORE TUNING / NO MORE TEST ACCESS FOR MODEL DEVELOPMENT

1. Executive Decision

Experiment B 已完成科學層面的封箱。 正式 Linear Experiment B 的第一次 Final Test 為 Negative Transfer；其後的 Experiment B2 為已揭露 Test 後之 Supplementary / Exploratory tuning，選定 B2-P1 後在相同 Plant1 Test 上呈現 Supplementary Positive Transfer。兩者必須永久分開記錄。

2. Data / Evaluation Contract

Item	Locked value
Direction	Source = Plant2; Target = Plant1
Sampling	15 minutes
Features	TIME_SIN, TIME_COS, IRRADIATION, AMBIENT_TEMPERATURE, MODULE_TEMPERATURE
Target	DC_POWER
Window / Horizon	5 / 1
Source rows	Train 2088 / Val 523 / Test 653
Target rows	Train 521 / Val 131 / Test 2612
Target sequences	Train 516 / Val 126 / Test 2607
Scaler	Feature scaler and target scaler separated; fit on Training only
Evaluation	Target inverse transform -> original-scale MAE / MSE / RMSE / R ²

3. Formal Experiment B

- Protocol: solar-linear-v1.0; Output activation = Linear.
- Formal run: 20260821T041718Z_seed1234.
- Validation selection was locked before one-time Target Test access.
- Formal B must remain preserved as the historical confirmatory run.

Method	MAE	MSE	RMSE	R ²	Classification
WOTL	28,769.9078	1,392,464,754.3387	37,315.7441	0.820814	Baseline
Formal PFT	36,009.4171	2,149,426,000.9080	46,361.9025	0.723406	Negative Transfer

4. Experiment B2 - Optimized Supplementary Strategy

- Run: 20260822T120239Z_seed1234.
- Selected candidate: B2-P1 = fine-tune Last LSTM + target-specific output head.
- Partial FT: trainable layers 4 and 6; earlier transferred layers and BatchNormalization frozen.
- Learning rate = 1e-5; batch = 128; max epochs = 500; seed = 1234; output = Linear.
- B2 does NOT replace Formal B and is NOT a new untouched Final Test.

Method	MAE	MSE	RMSE	R ²
WOTL	28,769.9078	1,392,464,754.3387	37,315.7441	0.820814
B2-P1	12,964.0142	550,250,743.1286	23,457.4241	0.929192

Numerical result: MAE, MSE and RMSE all decreased and R² increased; therefore B2-P1 is classified as Supplementary Positive Transfer.

5. Diagnostic / Governance Notes

- Predictions and metrics were rechecked on 2,607 aligned Test sequences; timestamp and y_true alignment passed.
- B2-P1 has 610 negative predictions versus WOTL 21 because Linear output does not impose a non-negative physical constraint.
- Official evaluation remains raw / unclipped; no post-hoc clipping is used to improve metrics.
- Single seed / single split: do not claim statistical significance, stable improvement, universal effectiveness or proven-best status.
- Allowed after sealing: read-only audit, thesis writing, descriptive analysis of existing predictions, SHA/provenance verification and oral-defense preparation.

6. Closure Status

Formal B	Preserved / sealed; Negative Transfer
B2-P1	Selected optimized supplementary strategy
B2 numerical classification	Supplementary Positive Transfer
Scientific closure	PASS
Repository archival closure	Pending LOCAL/Codex read-only inventory